

recent advancements have propelled it into the spotlight, reshaping the way vehicles interact and enhancing safety on the road.

V2V technology comes in two primary forms: ITS-G5 or DSRC, predominantly used in the European Union, and cellular V2X (vehicle-to-everything), favoured in countries such as South Korea, China, and the USA. These systems enable vehicles to communicate critical safety information directly to one another, using a dedicated part of the electromagnetic spectrum at 5.9GHz. This dedicated frequency ensures reliable and uncongested communication, setting the stage for a safer driving experience.

The key advantage of V2V communication over traditional cellular networks lies in its reliability, particularly in densely populated areas or during events that strain cellular networks. This dedicated communication channel guarantees the swift and consistent transmission of safety messages, making it an invaluable asset in emergency situations.

The scope of V2X technology extends beyond V2V communication; it encompasses various facets, including vehicle-to-infrastructure (V2I), vehicle-to-network (V2N), and the pivotal vehicle-to-vehicle (V2V) communication. What sets these systems apart is their ability to function independently of local cellular networks. Even in areas with limited or no 4G or 5G coverage, such as tunnels or remote locations, V2X technology remains operational. It empowers vehicles to 'see' through physical obstructions, enhancing reaction times and overall situational awareness, a boon for the emerging field of autonomous vehicles.

However, the adoption and protocols of V2V communication differ across the globe. Europe

initially favoured DSRC, a Wi-Fi-based technology. Still, there is a noticeable shift worldwide towards the newer CV2X technology, which employs the language of cellular networks for communication. The United States, China, and South Korea have already embraced CV2X, and Europe is expected to follow suit due to economic considerations and the availability of compatible hardware.

In regions like India and Japan, regulations and preferences surrounding V2V technology are still evolving. India displays a growing inclination towards CV2X. Meanwhile, Japan presents a unique scenario, with diverse technologies utilising different segments of the electromagnetic spectrum for emergency services.

How software in cars helps making money

The SDV market is rapidly transforming, with automakers increasingly integrating SDV features for monetisation. This growth is expected to contribute one-fifth of automotive revenue by 2034, driven by shifts towards 'autonomy as a service' and changing vehicle ownership models, favouring leasing and monthly software subscriptions. The industry anticipates average subscription revenue of about \$70 per month per vehicle, potentially generating \$700 billion annually, although hardware sales still constitute 80% of total revenue.

Globally, automakers are moving towards uniform pricing for software features. This consistency ensures standardised pricing across regions. Additionally, the ongoing revenue potential from subscriptions over a vehicle's lifespan, typically 12 years, presents a lucrative opportunity for automakers.

India, with its rapidly expanding 4G network coverage, presents an

intriguing market for software defined vehicles. Approximately 90% of India's population enjoys 4G connectivity, laying a strong foundation for the adoption of software-defined features. However, barriers such as import tariffs for electric vehicles have hindered the entry of cutting-edge technology. Nevertheless, if these obstacles are overcome, India could witness significant growth in the realm of software-defined vehicles, capitalising on its robust 4G network infrastructure.

Consumer sentiment plays a pivotal role in the success of subscription-based software services in vehicles. The way OEMs package these services will determine their attractiveness to consumers. Recent examples, like BMW's trial of heated steering wheel subscriptions, demonstrate the potential impact of negative consumer sentiment. When handled poorly, such subscription models can backfire, prompting automakers to rethink their strategies.

The automotive industry is undergoing a profound shift as SDVs become increasingly prevalent, reshaping the way we interact with cars. From basic navigation to advanced AI for autonomous driving, software plays a pivotal role in enhancing user experiences. Over-the-air updates offer convenience and faster innovation, while in-vehicle payments turn cars into payment devices, and AI-driven digital assistants make interactions more natural. V2V communication technology enhances safety, and the SDV market is poised to contribute significantly to automotive revenue. As the industry adapts to these changes, it promises safer, more convenient, and personalised driving experiences for consumers. **EFY**

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